

Key Findings discovered:

1A. Objective 1: Patient demography in relation to Death

The highest mortality rates are concentrated in the Senior (65-79) and Middle-Aged (45-64) groups, specifically among those in the "Basic" occupation category, such as hawkers and small-scale sellers.

The data reveals a critical trend: while Seniors in this category face a high mortality rate of 22.55%, Adults (25-44) in the exact same occupation group show a nearly identical risk level at 21.71%.

1B. Objective 2: Patient demography in relation to DAMA

The highest DAMA rates are concentrated in the Middle-Aged (45-64) and Adult (25-44) "Basic" occupation groups, recorded at 78.95% and 66.67% respectively. In contrast, Seniors (65-79) in the same occupation category show a significantly lower DAMA rate of 25.42%.

2. Objective 2: Reasons for DAMA

Financial factors are the overwhelming cause of Discharge Against Medical Advice, accounting for 87.68% of all recorded DAMA cases. When combining "Financial Constraint" and "Financial Incapability," the total reaches 242 out of 275 instances, surpassing all other reasons.

3. Objective 3: Prevalence and impact of chronic illness amongst patients

The study identifies Stroke (23.48%) and chronic kidney disease (21.92%) as the two most prevalent conditions. However, Diabetes recorded the highest mortality rate at 30.72%, followed by CKD (22.84%) and Stroke (21.90%).

When analyzed across demographics, both prevalence and mortality for these conditions are most heavily concentrated within the Middle-Aged (45-64) and Senior (65-79) age groups.

4. Impact of lifestyle factors on hospital outcomes.

NSAID use is the lifestyle factor with the highest mortality rate at 16.35% (34 deaths out of 208 cases). While death rates for all three factors are similar (15% to 16%), the high volume of NSAID cases makes it the most widespread risk in this study.

5. Evaluation of the performance of the doctors based on specialization, and workload.

Higher mortality is linked to doctors with the highest workloads (Dr. Ogbu in Emergency Medicine and Dr. Ogedegbe in Cardiology), suggesting pressure may be affecting patient outcomes.

General Surgery (Dr. Mahmoud) shows extremely high mortality (80%) despite low volume, indicating possible late presentations or case severity.

Nephrology is also imbalanced, with workload concentrated on Dr. Solanke despite the high prevalence and mortality rate of CKD.

Mortality remains concentrated in Emergency, Surgery, and Cardiology, while Orthopedics and Dermatology show lower risk.

RECOMMENDATIONS

1. TARGET FINANCIAL-DRIVEN DAMA WITH STRUCTURED PATIENT SUPPORT

The data clearly shows that DAMA is heavily driven by financial constraints, especially among middle-aged and senior patients in basic occupations such as (Traders, hawkers, and daily earners).

The hospital should introduce **financial counselling and social support systems**, such as:

Dedicated medical social workers

Flexible payment plans or staged billing so as to reduce the pressure of huge bills on patients

Early identification of high-risk patients (low-income groups)

This allows intervention before patients decide to leave.

2. ALIGN RESOURCE ALLOCATION WITH DISEASE BURDEN

With Stroke and CKD being the most prevalent conditions, the hospital should:

Prioritize drug availability and treatment resources for CKD and Stroke

Ensure critical units are properly equipped and staffed

Regularly review allocation based on case volume and mortality trends

This ensures the system is responding to where the real burden exists.

3. INTENSIFY PREVENTION AND MONITORING FOR HIGH-RISK CONDITIONS

Diabetes, CKD, and Stroke are driving mortality and cost.

The hospital should:

Strengthen **routine monitoring** for diabetic and hypertensive patients since hypertension is a major driver of this conditions

Increase patient education on lifestyle prevention

Promote early detection to reduce late, severe presentations

This reduces both mortality and long-term treatment cost.

4. REASSESS WORKLOAD DISTRIBUTION AND HIGH-MORTALITY UNITS

There is a clear pattern linking high workload and mortality, particularly in Emergency, Cardiology, and Nephrology.

The hospital should:

Investigate workload imbalance (e.g., CKD concentration on one doctor)

Redistribute cases or increase staffing where needed

Audit high-mortality units like General Surgery for case severity vs process gaps

This ensures patient outcomes are not compromised by operational strain.

LIMITATION

A significant portion of the data for the education category was not recorded.

This limits the ability to fully assess patient understanding, which is important for evaluating awareness of preventive healthcare practices and adherence to treatment.